The DICTO OCD Stella

The Ogdoecontahedron's companion cut along its joints: a golden solid that fills 151/2 − 267√5/8 ≈ 87.12% of space in the Ogdoecontahedron's packing, copies meeting face to face, and with regular octahedra 1423/32 − 3113√5/160 ≈ 96.33%

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Version 3 · 20 October 2026 (adds the crystals, exact: with octahedra and the notched OCD Stella, with icosahedra and the gap pieces; and the Atom 13) · DICTO DOI: 10.5281/zenodo.23430870 (all versions) · Licence CC BY-SA 4.0 · An introduction to one solid; its parent paper: The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688. Data and checks are included with this record.
The companion ("a free gift from the way the adapters sat on the OCD"), the stella cut, the seed, the name, the partners on the second network (icosahedra and octahedra), the pyrochlore arrangement, cutting the gap into pieces and the Atom 13 are DICTO's; the exact computations and the text were done with an AI assistant (Claude, Anthropic).

Abstract. The DICTO Ogdoecontahedron (OCD) packs in a body-centred cubic arrangement, each copy joined to 8 neighbours on hexagons. The golden adapter pieces around one copy have an icosahedral hull, the OCD's companion C (72 corners: 30 hexagons in golden planes and 20 triangles). Cutting C along the 8 joints gives the DICTO OCD Stella S: 48 corners and 50 faces, golden except for the joint triangles, which are exactly the packing cell's hexagon walls. Copies of S meet face to face across all 8 joints and fill exactly 151/2 − 267√5/8 ≈ 87.12% of space. Twelve OCD Stellas around one small one fill an outer OCD Stella exactly at φ^2, the Bodhi seed's pattern, though S is only pyritohedral. With a partner on the second network of the packing the OCD Stella forms crystals: with the icosahedron of the OCD's hexagon planes, all face to face, 1523/32 − 669√5/32 ≈ 84.60%, the gap falling into four kinds of convex piece; with the regular octahedron of the eight joint planes, whose six tips nest in the OCD Stellas, 1423/32 − 3113√5/160 ≈ 96.33% (octahedra on one diamond network) or 3839/64 − 8453√5/320 ≈ 91.72% (rock salt), each Stella notched six times. Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

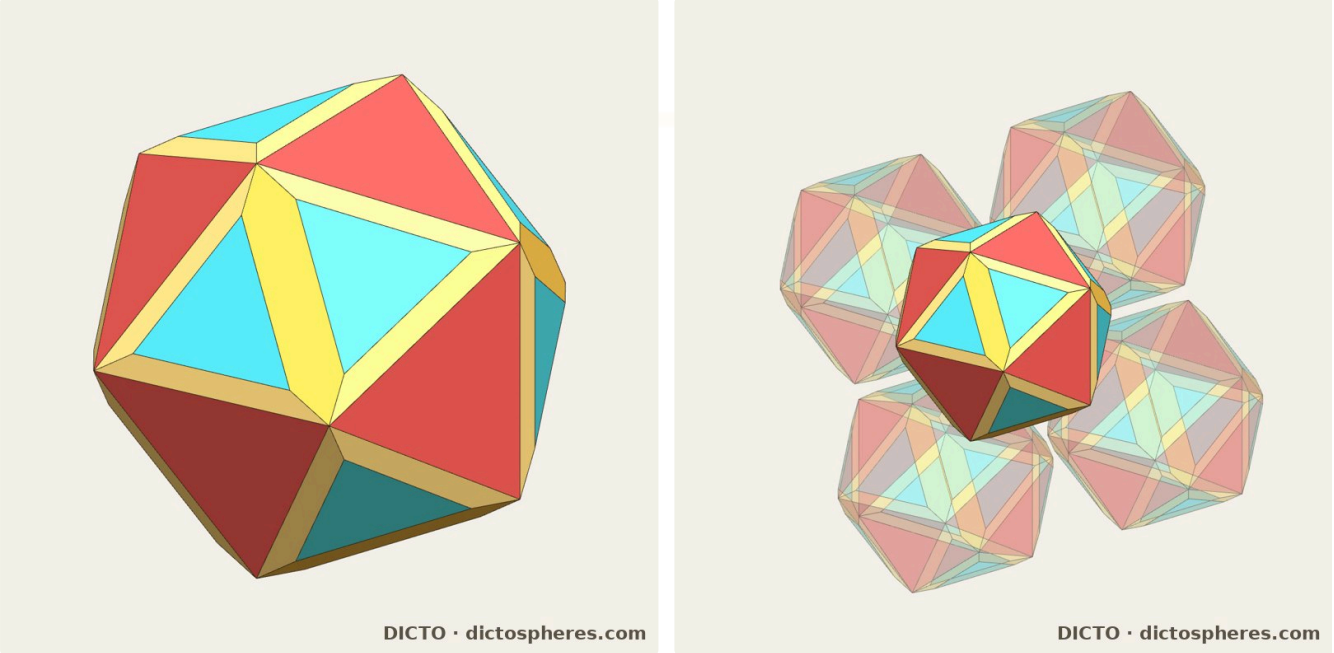


Figure 1. Left: the DICTO OCD Stella. Right: copies in the OCD's packing, meeting face to face across the joints.

1. The solid

Around one OCD of its packing (cell volume 512/729 + 256√5/729), the 60 golden adapter pieces (the full orbit of the 24 per cell) have a convex hull C with 72 corners, 30 hexagons in the golden planes of the OCD's envelope and 20 triangles in the three-fold planes through the adapters' outer corners (squared distance 14/27 − 10√5/243), volume (−18880 + 9760√5)/2187. Its combinatorial type is Conway t3jD, the triacontahedron truncated at its 20 three-fold corners, here at the depth set by the OCD (Antiprism). The DICTO OCD Stella is S = C cut by the packing's Voronoi cell, that is, along the 8 joints.

Property	Value (exact)
corners, faces	48 corners; 50 faces: 6 golden hexagons, 24 golden quadrilaterals, 12 triangles of C, 8 triangles in the OCD's joint hexagon planes (the cell's hexagon walls)
volume	(−4064 + 2240√5)/729 ≈ 1.296011
packing	fills 151/2 − 267√5/8 ≈ 87.12% of space in the OCD's packing; copies meet face to face across the 8 joints
symmetry	pyritohedral (the cut keeps the 24 symmetries the OCD shares with the cubic packing)

Table 1. The DICTO OCD Stella (`stella_cut_exact.py` → `stella-cut-exact.txt`).

2. The seed

The seed (exact). An outer OCD Stella of scale φ² holds 12 OCD Stellas on its five-fold axes, each touching it, and one of scale 1/φ at the centre touching all 12. By the Minkowski argument the seed closes exactly when no two of the twelve overlap; of the 66 pairs, 30 have gauge exactly 1 (touching, each copy 5 others, along the icosahedron's edges) and 36 exactly φ. So it closes exactly at φ² with the centre at 1/φ, and the gap is −69 + 31√5 ≈ 31.8%: the pattern of the Bodhi Seed (DICTO DOI: 10.5281/zenodo.23311353), though the OCD Stella is only pyritohedral (`stella_seed_exact.py`).

3. The crystals

The OCD's packing (sites on a body-centred cubic lattice, cell volume (512 + 256√5)/729) splits into two diamond networks, and DICTO proposed putting a partner on the second while the OCD Stellas stay on the first. Each partner below lies in its own cell, as the OCD Stella does, and meets its eight joint neighbours on the cell's hexagon walls.

Crystal	Fill (exact)	Space group (float)
OCD Stellas alone (both networks)	151/2 − 267√5/8 ≈ 87.12%	Im-3 (No. 204)
OCD Stellas + icosahedra (pyrochlore)	1523/32 − 669√5/32 ≈ 84.60%	Fd-3 (No. 203)
OCD Stellas + octahedra (pyrochlore)	1423/32 − 3113√5/160 ≈ 96.33%	Fd-3 (No. 203)
OCD Stellas + octahedra (rock salt, octahedra on a quarter of the sites)	3839/64 − 8453√5/320 ≈ 91.72%	Fm-3 (No. 202)

Table 2. The crystals. With icosahedra or octahedra on the pyrochlore networks it is the same crystal with the partner swapped: both solids on sites of symmetry 23 (spglib; float).

With icosahedra (exact). On every site of the second network sits the icosahedron of the OCD's own 20 hexagon planes, volume (−80 + 80√5)/81, its 12 corners on the cell's walls.

Every solid meets its 8 neighbours face to face, and the crystal fills 1523/32 − 669√5/32 ≈ 84.60%. DICTO proposed cutting the gap into pieces: cut by cones from each centre over its free faces, it falls into exactly four kinds of convex piece, 12 wedges round each icosahedron, (308 − 116√5)/2187 (6 corners, 5 faces), and round each OCD Stella 12 caps on its triangles, (346204 − 143140√5)/2381643 (9 corners, 7 faces), 6 slabs on its golden hexagons, (39896 − 17240√5)/216513 (16 corners, 12 faces), and 24 thin wedges on its golden quadrilaterals, (12596 − 5596√5)/88209 (6 corners, 5 faces). They sum exactly to the cells less the solids; the icosahedra's share of the gap is (961 + 209√5)/2454 ≈ 58.2%.

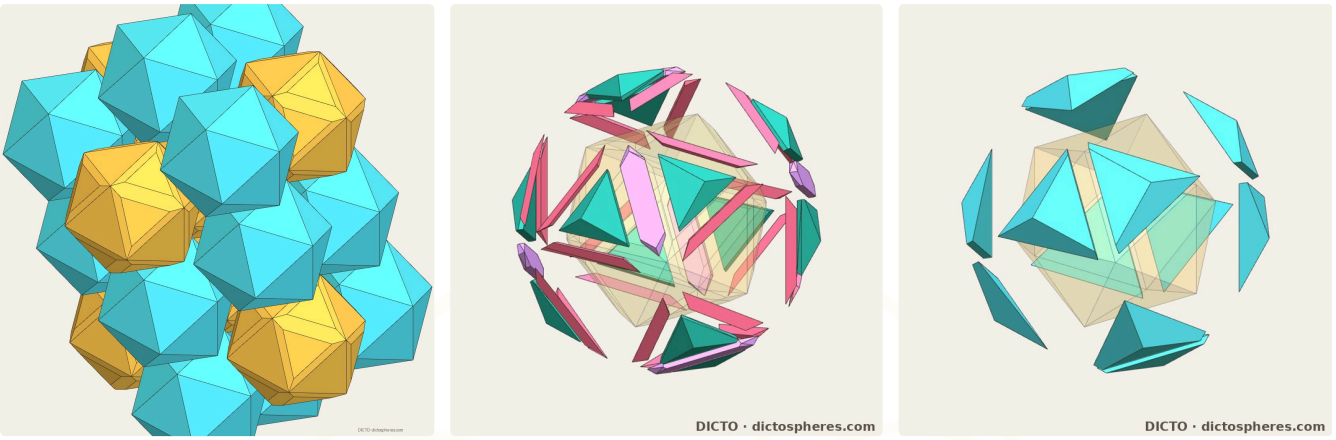


Figure 2. Left: OCD Stellas (gold) and icosahedra (blue) on the two networks, face to face. Middle: the gap round one OCD Stella in caps, slabs and thin wedges. Right: the 12 wedges round one icosahedron.

With octahedra (exact). DICTO proposed the octahedron next. The eight joint planes bound a regular octahedron, |x| + |y| + |z| ≤ R with R = 2φ/3 in cube axes, exactly 9/8 of the cell: it covers its whole cell, and its six tips pass the square walls, each (32 + 16√5)/2187, 1/48 of a cell, each apex halfway from the wall to the next site. Only the six square neighbours' tips reach an OCD Stella, each taking the same **notch**, 392/2187 − 88√5/1215 ≈ 0.017287 (6 notches: 8.00% of the Stella), leaving the **notched OCD Stella**, −1616/243 + 12784√5/3645 ≈ 1.192289. Octahedra on hexagon neighbours meet face to face; on square neighbours they would overlap tip to tip, so no two octahedra may be square neighbours. A square step changes a site's class within its parity, so with octahedra on one diamond network (one even and one odd class) every OCD Stella has octahedra on all six square sides: the crystal fills 1423/32 − 3113√5/160 ≈ 96.33%. With octahedra on one class only (rock salt) it fills 3839/64 − 8453√5/320 ≈ 91.72%. The float values found on 16 October agree.

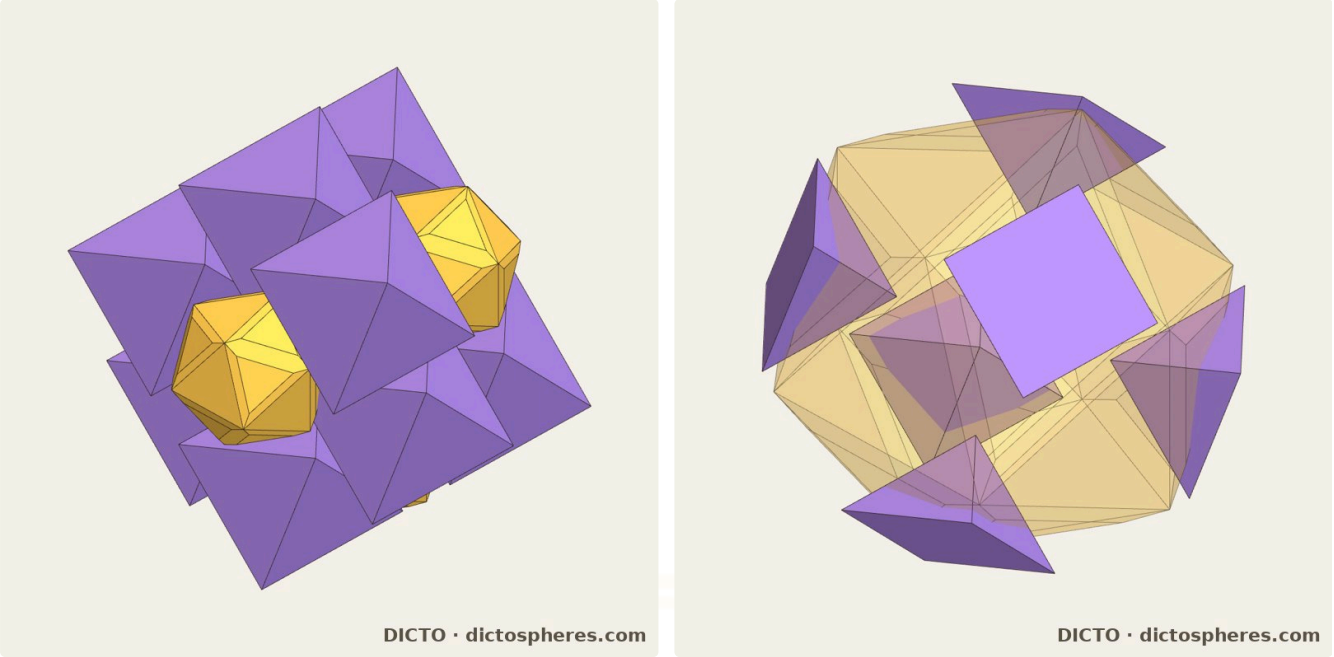


Figure 3. Left: octahedra (violet) and OCD Stellas (gold) on the pyrochlore networks. Right: one OCD Stella (glass) with the six octahedron tips in its notches.

The Atom 13 (float; DICTO proposed it). Twelve OCD Stellas on the five-fold axes of a central one, each pushed out until it touches the centre (distance 1.56743), meet the centre flat with six corners on its face planes and never touch each other (gap 0.31); the thirteen fill 49.82% of their outline. Set flat on the centre's twelve triangles instead, neighbours 41.8° apart overlap. Twenty icosahedra on the centre's twenty triangles, in the three-fold gaps between the twelve, fit up to 0.4878 of the crystal's icosahedron, limited by the twelve OCD Stellas; the 33 solids fill 58.20% of their outline.

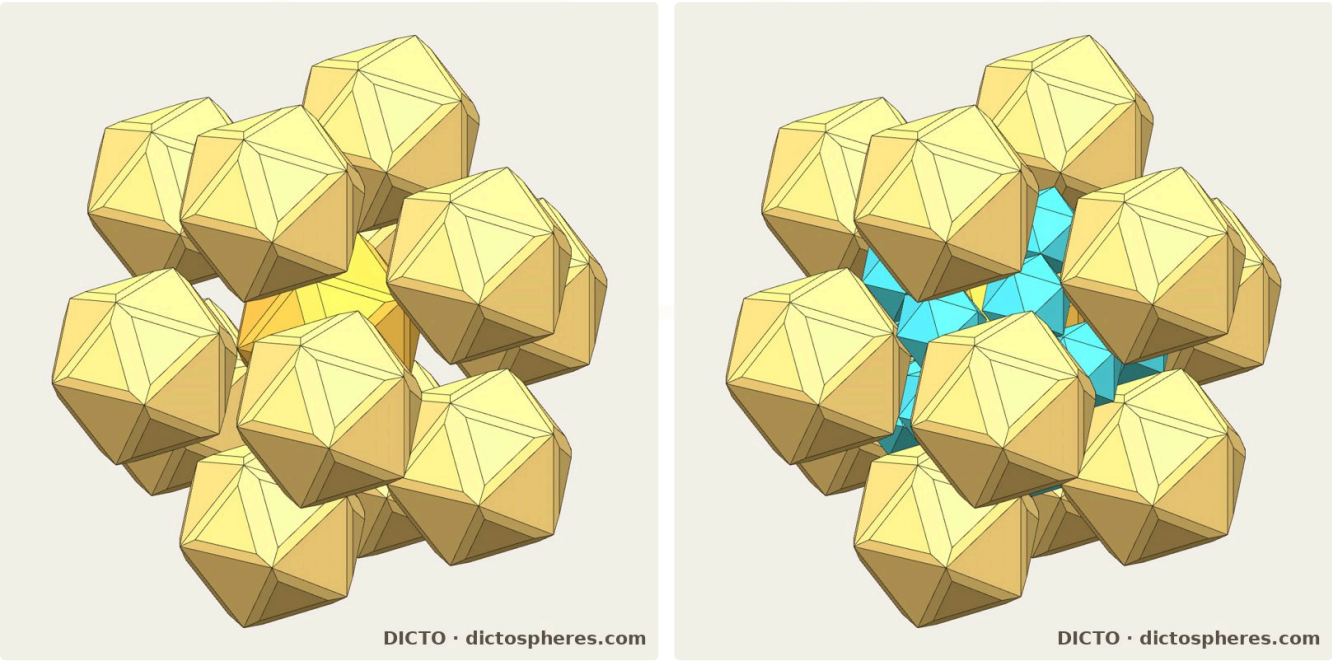


Figure 4. The DICTO OCD Stella Atom 13: left, thirteen OCD Stellas; right, with twenty icosahedra in the three-fold gaps.

4. Status

Known: the combinatorial type of C (Conway t3jD, the chamfered icosahedron) and a related golden space-filler cut from the rhombic triacontahedron on a body-centred cubic lattice (Kabai, below); binary crystals of hard polyhedra, such as tetrahedra with octahedra (Cadotte et al., Soft Matter 12, 7073–7078, 2016), and dense packings of the Platonic and Archimedean solids (Torquato and Jiao, Phys. Rev. E 80, 041104, 2009). **Not found, in the sources checked:** the OCD Stella, its 87.12% packing, and its seed; its crystals with icosahedra or octahedra, their fills, the gap pieces, the notched OCD Stella and the Atom 13 (library and web, 20 October 2026). Every "not found" is limited to the sources listed here and in the Appendix.

Sources checked on the web (17 October 2026):

- **Related and known: S. Kabai, "54-Faced Space-Filling Polyhedron"** (Wolfram Demonstrations Project): a rhombic triacontahedron with 8 half oblate golden rhombohedra removed at the corners of an inscribed cube, 6 golden rhombi and 48 triangles, filling space with its centres on a body-centred cubic lattice. A golden solid cut from the triacontahedron to pack on bcc is therefore known; the OCD Stella is a different solid (48 corners, 50 faces with golden hexagons and quadrilaterals, cut by the packing's Voronoi cell) and fills 87.12%, not all, of space.
- **Wikipedia**, "Chamfer (geometry)": the chamfered icosahedron (tritruncated rhombic triacontahedron), its equilateral form; no packing and no cut by a cubic cell.
- **Klitzing**, "chike" (the truncated rhombic triacontahedron); **McCooey**, Chamfered Solids (no chamfered-icosahedron packing).
- **Web search** for the packing fraction $151/2 - 267\sqrt{5}/8 \approx 87.12\%$ and for the name: no results.

Polyhedron software libraries (17 October 2026):

- **SageMath** polytope library (`polytopes.*`): its icosahedral constructors are the icosahedron, dodecahedron, icosidodecahedron, truncated icosahedron and the other Archimedean solids, and the pentakis dodecahedron; none of the solids here.
- **netlib polyhedra database** (142 entries: Platonic, Archimedean and their duals, prisms, the 92 Johnson solids and five others): none of the solids here (the pentakis dodecahedron, file 41, is the canonical one).
- **Antiprism** 0.29 (A. Rossiter): used to recompute and check the structures independently (for Vajrayana and Progenitors, and the companion's type t3jD); its Conway-operator sweep is a generator, not a catalogue.
- **Not run:** Wolfram `PolyhedronData` (needs Mathematica) and Stella / Great Stella (R. Webb; its library beyond the uniform, Johnson and Stewart-toroid models is not listed online). No claim is made about them.

The reference library: see the Appendix.

5. Files

In the study (branch research: studies/2026-10-16-ocd-pyrochlore): `stella_cut_exact.py` → `stella-cut-exact.txt`, `stella_seed_exact.py` → `stella-seed-exact.txt`, `ocd_gbpc_hull.py`, `companion_variants.py`; the crystals: `pyro_icosahedra.py`, `crystal_gap_exact.py` → `crystal-gap-exact.txt`, `stella_octa_pyro.py`, `crystal_symmetry.py` → `crystal-symmetry.txt`, `stella_atom13.py` →

`stella-atom13.txt`; and in studies/2026-10-20-stella-octahedra: `stella_octa_exact.py` → `stella-octa-exact.log` (the octahedra crystals, exact, computed in cube axes; volumes and fills do not depend on the turn). See it in the app: Kaleidohedra, 3D+ → OCD Stella Crystals.

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 234 entries. The full text of the 64 entries below was searched on 17 October 2026 for the solid's construction, its key numbers and its name (pentakis / kis-dodecahedron / hexecontahedron / hull of a dodecahedron and an icosahedron; enneacontahedron / rectified truncated icosahedron / compounds of the truncated icosahedron; chamfered or truncated triacontahedra; the four DICTO names). None contains any of the four solids introduced in these papers; the only hits are mentions of the classical pentakis dodecahedron (Kepler 1619; Messer 1995; de Graaf, van Roij and Dijkstra 2011) and of the chamfer operation (Goldberg 1937; the Goldberg-polyhedron and Fullerene articles). Not text-searched: 12 image-only scans and 158 link-only entries (copyrighted books and articles held as citations); no claim is made about them.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1619	Harmonices mundi libri V
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional en
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric

K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron